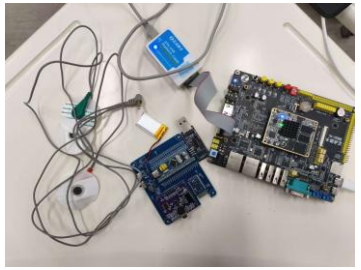
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# 基于FPGA的心电边缘分析系统

**Zynq-7020 · CNN硬件加速 · 逐拍心电分析**

刘源 · 王兴维 · 曾博文 | 北京邮电大学

同板同精度对照 | ARM / FPGA | 72个有效心拍



系统实物全景 · Zynq-7020

## 项目概览

面向便携式心电分析，构建无线采集、Zynq逐拍推理与网页记录链路。

- 波形显示
- 心率与逐拍分类
- 记录保存与CSV回放




网页回放示例 · 12个有效心拍

网页显示心率、逐拍类别与模型概率，并支持记录导出。

## 系统架构与软硬件分工

ADS1292R 500 Hz → STM32 + JDY-18 → 电脑 USB-UART

**Zynq-7020 · ARM + FPGA PL**

**ARM**

滤波、R峰检测、RR
切拍与特征准备
软件推理对照

↔

**FPGA PL**

CNN全网络推理
FP32 / W8A8
独立HGB分类核心

### 硬件加速实现

- 16路卷积输出并行
- 4路全连接输出并行
- LUTRAM权重存储
- W8A8混合精度 INT8 W/A · INT32累加 / 部分FP32

同一心拍、同一精度：核对输出后分别计时

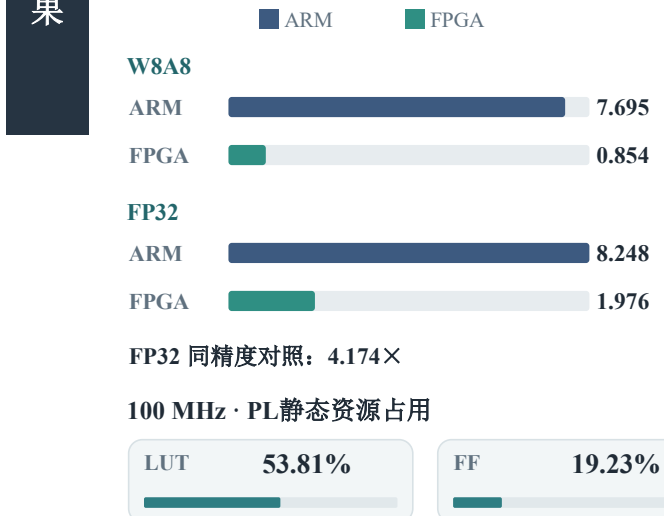
## 创新设计

## 实测结果

# 9.015×

**W8A8 CNN · ARM / FPGA 加速比**

每拍平均推理调用耗时（ms）




**FP32 同精度对照：4.174×**

**100 MHz · PL静态资源占用**

| | |
|---|---|
| LUT | 53.81% |
| FF | 19.23% |
| BRAM | 87.50% |
| DSP | 47.73% |

60秒合成输入，500 Hz；73个事件，72个有效拍。同板同精度、同H27权重。

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# FPGA ECG Edge Analysis System

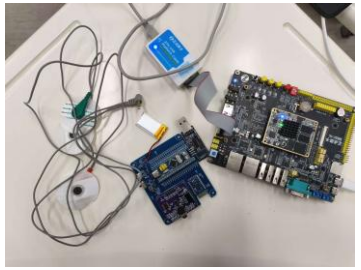
**Zynq-7020 · CNN hardware acceleration · Beat-level ECG analysis**

Yuan Liu · Xingwei Wang · Bowen Zeng | Beijing University of Posts and Telecommunications

Same board & precision

ARM / FPGA

72 valid beats



Complete system prototype · Zynq-7020

## OVERVIEW

A wireless ECG pipeline linking beat-level Zynq inference with web-based review and recording.

Waveform display

Heart rate & beat labels

Records & CSV replay




**Web replay example · 12 valid beats**

The web UI shows heart rate, beat labels and model scores, with record export.

## SYSTEM ARCHITECTURE & HW/SW PARTITION

ADS1292R 500 Hz → STM32 + JDY-18 → PC USB-UART

**Zynq-7020 · ARM + FPGA PL**

**ARM**

Filtering, R-peak & RR
Beat segmentation / features
Software inference baseline

↔

**FPGA PL**

Full CNN inference
FP32 / W8A8
Separate HGB classifier

### FPGA ACCELERATION DESIGN

- 16-way convolution output parallelism
- 4-way dense output parallelism
- LUTRAM weight storage
- W8A8 mixed precision INT8 W/A · INT32 accum. + partial FP32

Same beat & precision: verify outputs, then time each path

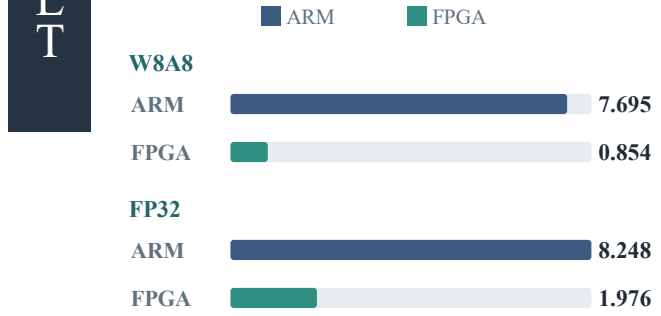
CREATIVE DESIGN

RESULT

# 9.015×

**W8A8 CNN · ARM-to-FPGA speedup**

Mean inference-call time per beat (ms)




**FP32 same-precision comparison: 4.174×**

**100 MHz · Post-route PL resource use**

| | |
|---|---|
| LUT | 53.81% |
| FF | 19.23% |
| BRAM | 87.50% |
| DSP | 47.73% |

60 s synthetic input at 500 Hz; 73 events, 72 valid beats. Same board, same precision and H27 weights.